Survival Models Tutorial 2: Solutions

1. The Kaplan-Meier estimate is obtained from the following table:

Interval	nj	dj	(nj-dj)/nj	S(t)
0-	10	0	1	1
2-	10	1	9/10	0.9
5-	9	1	8/9	0.8
8-	7	1	6/7	0.686
9-	6	1	5/6	0.571
12-	4	1	3/4	0.429
20-	1	1	0	0

where `S(t)` stands for $\hat{S}(t)$.

You can confirm this solution with the `survfit( )` function in R; this also produces the required plot. The code is on the web site. Note that the plot shows the 95% confidence interval and the position of the censored observations.

2. (i) Failures occur at times 3, 5, 10 and 12. The contributions to $L(\beta)$ are

$$\text{at 3: } \frac{1}{4+4e^\beta}; \quad \text{at 5: } \frac{e^\beta}{3+4e^\beta}; \quad \text{at 10: } \frac{1}{1+e^\beta} \quad \text{and at 12: } 1$$

$$\Rightarrow \quad L(\beta) \propto \frac{1}{4(1+e^\beta)} \times \frac{e^\beta}{3+4e^\beta} \times \frac{1}{1+e^\beta} \times 1 \propto \frac{e^\beta}{(1+e^\beta)^2(3+4e^\beta)}.$$

(ii) We have

$$\ell = \beta - 2\log(1+e^\beta) - \log(3+4e^\beta)$$

$$\frac{\partial \ell}{\partial \beta} = 1 - \frac{2e^\beta}{1+e^\beta} - \frac{4e^\beta}{3+4e^\beta} = 0 \quad \text{at the MLE}$$

$$\text{or} \quad (1+x)(3+4x) - 2x(3+4x) - 4x(1+x) = 0 \quad \text{where } e^\beta = x$$

$$\Rightarrow \quad \beta = \log\left(\frac{\sqrt{105}-3}{16}\right) = -0.792.$$

Differentiating $\partial\ell/\partial\beta$ we find

$$\frac{\partial^2 \ell}{\partial \beta^2} = -\frac{2e^\beta}{(1+e^\beta)^2} - \frac{12e^\beta}{(3+4e^\beta)^2} = -0.664 \quad \text{at } e^\beta = (\sqrt{105}-3)/16.$$

Hence $\hat{\beta} = -0.792$; the variance is $\mathrm{V}(\hat{\beta}) = 1/0.664$ and so the standard error is $\sqrt{1/0.664} = 1.23$.

This gives a test statistic

$$z = \frac{\hat{\beta} - \beta_0}{\sqrt{\mathrm{V}(\hat{\beta})}} = \frac{-0.792 - 0}{1.23} = -0.645$$

so $z^2 = 0.416$. Under the null hypothesis of the Wald test z comes from a $N(0,1)$ distribution (equivalently, z^2 comes from a χ^2_1 distribution).
A test statistic z or z^2 far from 0 would indicate a poor fit, so the p-value (calculated in R for example) is

$$\Pr(Z^2 \geq 0.416) = \Pr(|Z| \geq 0.645) = 2 \times \Pr(Z \leq -0.645) = 0.519$$

where $Z \sim N(0,1)$, so there is little evidence to support the alternative hypothesis that $\beta \neq 0$ (although of course the experiment is very small so will have low power).
Instead using the Tables (since R is not available in the exam), we see that for $Z^2 \sim \chi^2_1$,

$$\Pr(Z^2 \leq 0.4549) = 0.5$$

and so

$$\Pr(Z^2 \geq 0.416) = 1 - \Pr(Z^2 \leq 0.416) > 1 - \Pr(Z^2 \leq 0.4549) = 0.5$$

implying that there is no evidence to reject the null hypothesis.
Alternatively, from the Tables we see that $\Pr(Z^2 \leq 0.5) = 0.5205$ so that

$$\Pr(Z^2 \geq 0.416) > \Pr(Z^2 \geq 0.5) = 1 - 0.5205 = 0.4795.$$

(iii) We have

$$U(0) = 1 - \frac{2}{2} - \frac{4}{7} = -\frac{4}{7} \quad \text{and} \quad I(0) = \frac{2}{2^2} + \frac{12}{7^2} = \frac{73}{98}$$

and so the score test statistic is $S = U(0)^2/I(0) = 32/73 = 0.438$.
For $Z^2 \sim \chi^2_1$, we have

$$\Pr(Z^2 \geq 0.438) > \Pr(Z^2 \geq 0.5) = 1 - 0.5205 = 0.4795$$

implying insufficient evidence to reject the null hypothesis that $\beta = 0$.

(iv) Using (ii) we compute $\ell(0) = -3.332205$ and $\ell(\hat{\beta}) = -3.110237$ and so $-2\log\Lambda = -2(\ell(0) - \ell(\hat{\beta})) = 0.444$.
As above, for $Z^2 \sim \chi^2_1$, we have

$$\Pr(Z^2 \geq 0.444) > \Pr(Z^2 \geq 0.5) = 1 - 0.5205 = 0.4795$$

implying insufficient evidence to reject the null hypothesis that $\beta = 0$.

(v) You can confirm this solution with the `coxph( )` function in R. The code is on the web site.

The test statistics are consistent since $z^2 \approx S \approx -2\log\Lambda$.

3. (a) The partial likelihood is

$$\frac{1}{14+15e^{\beta}} \cdot \frac{e^{\beta}}{12+14e^{\beta}} \cdot \frac{e^{\beta}}{12+13e^{\beta}} \cdot \frac{1}{12+11e^{\beta}} \cdot \frac{1}{10+11e^{\beta}} \cdot \frac{1}{8+8e^{\beta}} \cdot \frac{1}{7+8e^{\beta}} \cdot \frac{1}{6+8e^{\beta}}.$$

so the partial log likelihood is (with $e^{\beta} = \phi$)

$$2\beta - \log(14+15\phi) - \log(12+14\phi) - \log(12+13\phi) - \log(12+11\phi) - \ldots$$
$$\ldots - \log(10+11\phi) - \log(8+8\phi) - \log(7+8\phi) - \log(6+8\phi).$$

(b) Here is the R-code to define the partial log likelihood. The full script is on the web site.

```
PartialL <- function(b){
  p <- exp(b)
  2*b - log(14+15*p) - log(12+14*p) - log(12+13*p)- log(12+11*p) -
      log(10+11*p) - log(8+8*p) - log(7+8*p) - log(6+8*p)
  }
```

Notice how we make life easier for ourselves be defining `exp(b)` as `p`.

(c) We use the R-function `PartialL( )` above to plot the partial log likelihood. We easily find that $\hat{\beta} = -1.19$; see the web site for the complete code.

(d) We define the survival times, the censoring indicator and the group factor in the usual way and then fit the Cox model with

```
Cox.fit <- coxph(Surv(Times, Censor) ~ factor(Treat), method = "breslow")
```

Then `summary(Cox.fit)` gives us everything we want. We confirm the estimate of β as -1.19; additionally we learn that the standard error of $\hat{\beta}$ is 0.817, the LRT statistic is 2.48, $z^2 = 2.13$ (the Wald test) and the score test is 2.39. These statistics have p-values of around 0.12 - weak evidence against the null hypothesis of equal forces of mortality for the two groups. Again the complete code is on the web site.

4. (a) There are claims at times 3, 5, 9, 10, 15, and 16 so the partial likelihood is

$$\begin{aligned} PL(\beta) &\propto \frac{1}{5+5e^{\beta}} \cdot \frac{e^{\beta}}{4+5e^{\beta}} \cdot \frac{e^{\beta}}{2+3e^{\beta}} \cdot \frac{1}{2+2e^{\beta}} \cdot \frac{1}{1+e^{\beta}} \\ &\propto \frac{e^{2\beta}}{(1+e^{\beta})^3(4+5e^{\beta})(2+3e^{\beta})} \end{aligned}$$

with partial log likelihood

$$P\ell(\beta) = 2\beta - 3\log(1+e^{\beta}) - \log(4+5e^{\beta}) - \log(2+3e^{\beta}).$$

The score function is

$$U(\beta) = 2 - \frac{3e^{\beta}}{1+e^{\beta}} - \frac{5e^{\beta}}{4+5e^{\beta}} - \frac{3e^{\beta}}{2+3e^{\beta}}.$$

(b) The R-code to define this partial log likelihood is

```
Partial <- function(b) {
  p <- exp(b)
  2*b - 3*log(1 + p) - log(4 + 5*p)- log(2 + 3*p)
}
```

The R code is on the Web site to define $U(\beta)$ as an R function.

(c) The plot follows and R tells us that $\hat{\beta} = -0.534$.

(d) Setting $\phi = \exp(\beta)$ in $PL(\beta)$ we find

$$\begin{aligned} PL(\phi) &= \frac{\phi^2}{(1+\phi)^3(4+5\phi)(2+3\phi)} \\ \Rightarrow P\ell(\phi) &= 2\log\phi - 3\log(1+\phi) - \log(4+5\phi) - \log(2+3\phi) \\ \Rightarrow U(\phi) &= \frac{2}{\phi} - \frac{3}{1+\phi} - \frac{5}{4+5\phi} - \frac{3}{2+3\phi}. \end{aligned}$$

(e) The Web site has the code and it tells us that $\hat{\phi} = 0.586$. You can either estimate the slope by reading some values off the graph or you can use this nice approximation

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \text{ for small } h,$$

where f is the score function U. We find that the standard error of $\hat{\phi}$ is about 0.54.

(f) $\exp(\hat{\beta}) = e^{-0.534} = 0.586 = \hat{\phi}$.

(g) Let's denote the partial log likelihood (a) wrt β by ℓ_{β} and (b) wrt ϕ by ℓ_{ϕ}. We need to check that the two values of "$-2\log\Lambda$" are equal, i.e.

$$-2\{\ell_{\beta}(0) - \ell_{\beta}(\hat{\beta})\} = -2\{\ell_{\phi}(1) - \ell_{\phi}(\hat{\phi})\}.$$

Using R we easily find that this is the case and both values are equal to 0.348 (as found by using `coxph( )`).